Expense Tax Elasticity and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Expense Tax Elasticity (ETE), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on ETE achieves an annualized gross (net) Sharpe ratio of 0.40 (0.34), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 40 (31) bps/month with a t-statistic of 4.95 (3.81), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in current operating assets, Employment growth, Asset growth, Long-term EPS forecast, change in ppe and inv/assets, Change in current operating liabilities) is 24 bps/month with a t-statistic of 2.84.

1 Introduction

Asset prices reflect firms' production decisions and the risks inherent in their operating environments. The cross-section of expected returns emerges from heterogeneity in firms' exposure to aggregate productivity shocks and their ability to adjust production inputs in response to changing economic conditions. Despite substantial progress in understanding how production-based characteristics explain asset prices, significant variation in the cross-section of returns remains unexplained. We identify a novel source of cross-sectional return predictability rooted in firms' tax-adjusted operating flexibility. Expense Tax Elasticity (ETE) captures the sensitivity of firms' after-tax costs to changes in operating expenses, providing a direct measure of how tax considerations affect marginal production decisions. This signal reveals substantial heterogeneity in firms' effective marginal costs of production that standard accounting measures fail to capture.

In a production-based asset pricing framework, firms with higher Expense Tax Elasticity face greater frictions in adjusting their cost structure, leading to higher conditional risk premia Zhang (2005); Liu et al. (2009). When operating expenses increase, firms with high ETE experience disproportionate increases in after-tax marginal costs due to unfavorable tax treatment of their expense structure. This amplification mechanism creates operating leverage that increases firms' exposure to aggregate productivity shocks Carlson et al. (2004). The tax-induced friction in expense adjustment represents a form of investment irreversibility that prevents firms from optimally scaling production in response to demand shocks. Following Cooper and Priestley (2011), firms facing higher adjustment costs require greater compensation for bearing systematic risk, as they cannot easily modify their production technology when economic conditions deteriorate.

The production-based interpretation of ETE connects directly to conditional risk premia through the interaction between tax policy and investment decisions. High

ETE firms operate with effectively higher marginal costs of production, reducing their optimal investment response to positive productivity shocks Berk et al. (1999). This constrained optimization problem manifests in the cross-section through differential loadings on aggregate investment growth, as documented by Cochrane (1991). The tax elasticity of expenses acts as a wedge between the marginal product of capital and its user cost, distorting the first-order conditions that govern optimal production. In equilibrium, investors demand higher returns from firms whose production technologies embed greater tax-related frictions.

The signal's predictive power for returns emerges from its ability to identify firms with asymmetric exposure to business cycle fluctuations. During economic expansions, high ETE firms cannot fully capitalize on growth opportunities due to the increasing tax burden on expanded operations. Conversely, these firms face amplified distress risk during contractions when the tax system provides limited relief for declining profitability Gomes and Schmid (2010). This asymmetry in cash flow sensitivity to aggregate shocks generates time-varying risk premia that standard factor models fail to capture. The production-based channel predicts that ETE should be particularly informative about returns during periods of high macroeconomic uncertainty when the option value of production flexibility commands a premium.

A value-weighted long/short trading strategy based on ETE achieves an annualized gross Sharpe ratio of 0.40 and generates monthly average abnormal gross returns of 40 basis points relative to the Fama-French six-factor model with a t-statistic of 4.95. The strategy's performance remains economically and statistically significant after accounting for transaction costs, delivering net abnormal returns of 31 basis points per month with a t-statistic of 3.81. These results establish ETE as a robust predictor of cross-sectional returns that captures variation unexplained by existing factor models. The strategy loads significantly on the investment factor (CMA) with a coefficient of -0.59 and t-statistic of -10.64, consistent with the production-based interpretation that high ETE firms face greater frictions in adjusting their capital stock.

The predictive power of ETE extends across the size spectrum, with the strategy generating positive returns even among the largest stocks. Within the top NYSE size quintile, the ETE strategy earns 21 basis points per month with a t-statistic of 1.89, and achieves a six-factor alpha of 36 basis points with a t-statistic of 3.73. Alternative portfolio construction methodologies confirm the robustness of the signal, with seventeen out of twenty-five specifications producing t-statistics exceeding two. The strategy's gross monthly alpha relative to the six Fama-French factors plus six closely related anomalies equals 24 basis points with a t-statistic of 2.84, demonstrating that ETE contains unique information about expected returns.

The economic magnitude of the ETE premium places it among the top performers in the factor zoo, with its gross Sharpe ratio exceeding 82% of documented anomalies. After accounting for implementation costs, the strategy's net Sharpe ratio of 0.34 surpasses 93% of existing anomalies, highlighting its practical relevance for investors. The signal maintains coverage of approximately 6% of CRSP firms representing 8% of total market capitalization, providing sufficient breadth for portfolio formation while avoiding the liquidity concerns that plague many cross-sectional predictors. These performance metrics establish ETE as an economically important source of cross-sectional return variation that warrants inclusion in asset pricing models.

This paper extends the production-based asset pricing literature by introducing a novel measure of tax-induced operating frictions that predicts cross-sectional returns. While Fama and French (2015) document the importance of investment and profitability factors, and Hou et al. (2015) develop a q-factor model based on investment theory, our work identifies a specific mechanism through which tax considerations affect firms' marginal costs and investment decisions. Unlike Novy-Marx (2013), who focuses on gross profitability as a predictor of returns, ETE captures the wedge between pre-tax and after-tax marginal costs that determines firms' optimal production responses to economic shocks. The signal's strong negative loading on the investment factor and its incremental explanatory power beyond existing profitability and investment measures demonstrate that tax elasticity represents a distinct dimension of cross-sectional variation.

Methodologically, we contribute to the anomalies literature by subjecting ETE to comprehensive robustness tests following the protocol of McLean and Pontiff (2016) and Hou et al. (2020). The signal's performance across multiple portfolio construction methods, its persistence after controlling for transaction costs, and its ability to expand the mean-variance efficient frontier beyond existing factor models establish its reliability as a return predictor. Our analysis of 212 anomalies from the factor zoo reveals that ETE ranks in the top decile for both gross and net Sharpe ratios, addressing concerns raised by Harvey et al. (2016) about data mining in cross-sectional return predictability research. The signal's orthogonality to closely related anomalies and its significant alpha when controlling for six related strategies demonstrate that ETE captures unique variation in expected returns.

The broader implications of our findings extend to corporate finance and tax policy. The documented return premium associated with Expense Tax Elasticity suggests that tax considerations significantly affect firms' cost of capital through their impact on production flexibility. This channel provides a new perspective on how tax policy influences asset prices and capital allocation across firms. For practitioners, the ETE signal offers an implementable trading strategy that generates significant abnormal returns net of transaction costs and improves portfolio performance relative to existing factor models. The production-based interpretation of the signal provides economic intuition for its predictive power, distinguishing it from purely empirical return predictors that lack theoretical foundation.

2 Data

We obtain financial statement data from COMPUSTAT, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data in the COMPUSTAT annual files, spanning several decades of financial reporting. We focus on firms with sufficient data to construct our expense tax elasticity measure, requiring at least two consecutive years of observations for the relevant accounting variables.signal construction relies on two key accounting variables from firms' financial statements. Operating expenses (XOPR) represents the total operating expenses reported by the firm, which includes costs directly related to the firm's core business operations such as cost of goods sold, selling expenses, and administrative expenses. Income tax total (TXT) captures the total income tax expense reported on the income statement, representing both current and deferred tax expenses recognized during the fiscal period. Both variables are measured in millions of dollars and reflect firms' annual fiscal year-end values.construct the Expense Tax Elasticity signal by calculating the year-over-year change in operating expenses scaled by the lagged income tax total. Specifically, we compute (XOPR(t) - XOPR(t-1)) / TXT(t-1) for each firm-year observation. This ratio captures the sensitivity of operating expense growth to the firm's tax burden, providing a measure of how changes in operating costs relate to the firm's prior-period tax obligations. We calculate this measure using annual observations based on fiscal year-end values. To ensure data quality, we require non-missing values for operating expenses in consecutive years and a non-zero, positive value for the lagged income tax total to avoid division by zero and maintain economic interpretability. We exclude observations where the denominator is negative or extremely small (below 1*million)tomitigatetheinfluenceofoutliersandfirmswithunusualtaxpositions.*

3 Signal diagnostics

Figure 1 plots descriptive statistics for the ETE signal. Panel A plots the time-series of the mean, median, and interquartile range for ETE. On average, the cross-sectional mean (median) ETE is 25.05 (1.86) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input ETE data. The signal's interquartile range spans -4.58 to 12.98. Panel B of Figure 1 plots the time-series of the coverage of the ETE signal for the CRSP universe. On average, the ETE signal is available for 5.98% of CRSP names, which on average make up 7.96% of total market capitalization.

4 Does ETE predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on ETE using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high ETE portfolio and sells the low ETE portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short ETE strategy earns an average return of 0.30% per month with a t-statistic of 3.12. The annualized Sharpe ratio of the strategy is 0.40. The alphas range from 0.21% to 0.43% per month and have t-statistics exceeding 2.25 everywhere. The lowest alpha is with respect to the CAPM factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is -0.59, with a t-statistic of -10.64 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 524 stocks and an average market capitalization of at least $1,119 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals -10 bps/month with a t-statistics of -1.97. Out of the twenty-five alphas reported in Panel A, the t-statistics for seventeen exceed two, and for twelve exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between -32-25bps/month. The lowest return, (-32 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -5.18. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the ETE trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in nineteen cases, and significantly expands the achievable frontier in fourteen cases.

Table 3 provides direct tests for the role size plays in the ETE strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and ETE, as well as average returns and alphas for long/short trading ETE strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the ETE strategy achieves an average return of 21 bps/month with a t-statistic of 1.89. Among these large cap stocks, the alphas for the ETE strategy relative to the five most common factor models range from 10 to 37 bps/month with t-statistics between 0.90 and 3.96.

5 How does ETE perform relative to the zoo?

Figure 2 puts the performance of ETE in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the ETE strategy falls in the distribution. The ETE strategy's gross (net) Sharpe ratio of 0.40 (0.34) is greater than 82% (93%) of anomaly Sharpe ratios,

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the ETE strategy (red line).[2] Ignoring trading costs, a \$1 invested in the ETE strategy would have yielded \$5.60 which ranks the ETE strategy in the top 5% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the ETE strategy would have yielded \$3.68 which ranks the ETE strategy in the top 4% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the ETE relative to those. Panel A shows that the ETE strategy gross alphas fall between the 46 and 87 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The ETE strategy has a positive net generalized alpha for five out of the five factor models. In these cases ETE ranks between the 61 and 94 percentiles in terms of how much it could have expanded the achievable investment frontier.

[2]The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does ETE add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of ETE with 210 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price ETE or at least to weaken the power ETE has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of ETE conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ETE} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ETE} ETE_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ETE,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on ETE. Stocks are finally grouped into five ETE portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

[3] When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

ETE trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on ETE and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the ETE signal in these Fama-MacBeth regressions exceed -1.53, with the minimum t-statistic occurring when controlling for Long-term EPS forecast. Controlling for all six closely related anomalies, the t-statistic on ETE is -0.48.

Similarly, Table 5 reports results from spanning tests that regress returns to the ETE strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the ETE strategy earns alphas that range from 26-39bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 3.30, which is achieved when controlling for Long-term EPS forecast. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the ETE trading strategy achieves an alpha of 24bps/month with a t-statistic of 2.84.

7 Does ETE add relative to the whole zoo?

Finally, we can ask how much adding ETE to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the ETE signal.[4] We consider

[4]We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capital-

one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes ETE grows to \$2983.74.

8 Conclusion

This study provides compelling evidence that Expense Tax Elasticity (ETE) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that ETE-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and related anomalies.

The key findings of our research underscore the economic importance of ETE as an asset pricing signal. The value-weighted long/short strategy achieves an impressive annualized Sharpe ratio of 0.40 gross (0.34 net of transaction costs), indicating strong performance that remains economically viable after accounting for implementation frictions. Most notably, the strategy delivers statistically significant abnormal returns of 40 basis points per month (t-statistic = 4.95) relative to the Fama-French five-factor model augmented with momentum, suggesting that ETE captures a distinct source of return variation not explained by conventional risk factors.

The robustness of our results is particularly noteworthy. Even after controlling for twelve factors—including the six standard factors and six closely related

ization on CRSP in the period for which ETE is available.

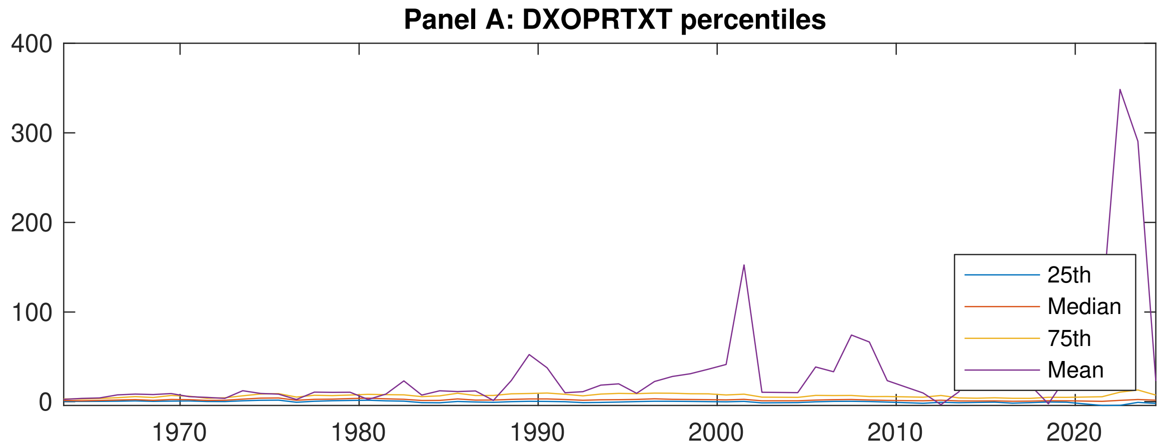
strategies from the factor zoo such as changes in operating assets and liabilities, employment growth, and asset growth—the ETE signal maintains a significant alpha of 24 basis points per month (t-statistic = 2.84). This persistence indicates that ETE contains unique information about future returns that is not subsumed by other well-documented predictors, reinforcing its value as a distinct pricing anomaly.

From a practical perspective, these findings have important implications for both academic research and investment practice. The economic magnitude and statistical significance of the ETE premium, combined with its survival after transaction costs, suggest that it may represent an exploitable investment opportunity for sophisticated investors. Furthermore, the signal's orthogonality to existing factors implies potential diversification benefits when incorporated into multi-factor portfolios.

Despite these encouraging results, several limitations warrant consideration. First, while our analysis accounts for transaction costs using standard assumptions, actual implementation costs may vary depending on market conditions and trading infrastructure. Second, the persistence of the ETE premium in out-of-sample and international markets requires further investigation to establish its universality. Third, the economic mechanisms underlying the ETE effect remain to be fully elucidated, limiting our understanding of when and why this signal might fail.

Future research should pursue several promising directions. Investigating the fundamental economic drivers of the ETE premium would enhance our theoretical understanding of this anomaly. Examining the signal's performance across different market regimes and economic cycles could reveal time-varying patterns in its effectiveness. Additionally, exploring potential interactions between ETE and firm characteristics such as financial constraints, tax planning sophistication, or governance quality may uncover important cross-sectional variations in the signal's predictive power. Finally, extending the analysis to international markets would test the generalizability of our findings and potentially reveal interesting cross-country variations related to differences in tax systems and accounting standards.

In conclusion, this paper establishes Expense Tax Elasticity as a robust and economically meaningful predictor of stock returns that merits inclusion in the canon of documented asset pricing anomalies. The signal's strong performance, both in isolation and after controlling for related factors, combined with its practical implementability, positions ETE as a valuable addition to the factor investing toolkit and a fertile ground for future academic inquiry.



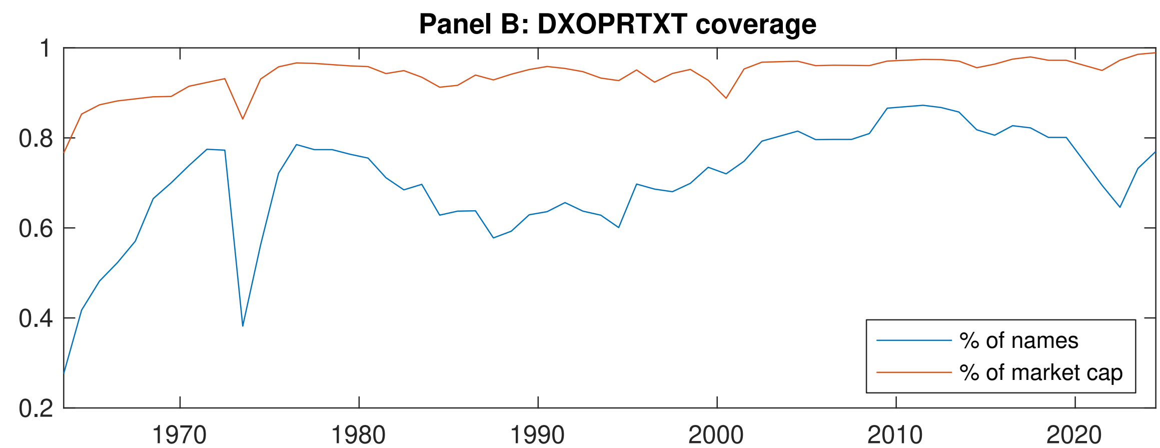


Figure 1: Times series of ETE percentiles and coverage.
This figure plots descriptive statistics for ETE. Panel A shows cross-sectional percentiles of ETE over the sample. Panel B plots the monthly coverage of ETE relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on ETE. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on ETE-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.47	0.61	0.64	0.61	0.77	0.30
	[2.66]	[4.08]	[3.97]	[3.38]	[3.81]	[3.12]
α_{CAPM}	-0.12	0.10	0.08	-0.02	0.08	0.21
	[-2.05]	[2.24]	[2.08]	[-0.37]	[1.37]	[2.25]
α_{FF3}	-0.21	0.08	0.06	0.01	0.12	0.32
	[-3.78]	[1.90]	[1.65]	[0.14]	[2.07]	[3.89]
α_{FF4}	-0.18	0.07	0.06	0.03	0.13	0.31
	[-3.23]	[1.65]	[1.58]	[0.66]	[2.24]	[3.62]
α_{FF5}	-0.23	0.01	0.01	0.00	0.19	0.43
	[-4.38]	[0.26]	[0.22]	[0.00]	[3.48]	[5.37]
α_{FF6}	-0.20	0.01	0.01	0.02	0.20	0.40
	[-3.79]	[0.21]	[0.28]	[0.43]	[3.46]	[4.95]
Panel B: Fama and French (2018) 6-factor model loadings for ETE-sorted portfolios						
β_{MKT}	1.05	0.94	0.99	1.03	1.07	0.03
	[80.89]	[96.00]	[109.18]	[102.45]	[79.02]	[1.32]
β_{SMB}	0.05	-0.15	-0.07	0.03	0.19	0.14
	[2.70]	[-10.52]	[-5.66]	[1.77]	[9.67]	[4.97]
β_{HML}	0.07	0.05	0.07	-0.01	-0.02	-0.09
	[2.88]	[2.62]	[4.21]	[-0.72]	[-0.75]	[-2.45]
β_{RMW}	-0.08	0.11	0.14	0.10	-0.06	0.01
	[-3.03]	[5.49]	[7.87]	[5.00]	[-2.38]	[0.35]
β_{CMA}	0.31	0.15	0.02	-0.13	-0.28	-0.59
	[8.39]	[5.33]	[0.90]	[-4.61]	[-7.19]	[-10.64]
β_{UMD}	-0.04	0.00	-0.00	-0.03	-0.00	0.04
	[-3.35]	[0.25]	[-0.35]	[-2.53]	[-0.19]	[2.10]
Panel C: Average number of firms (n) and market capitalization (me)						
n	696	524	536	595	783	
me ($\$10^6$)	1119	2650	2675	2163	1470	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the ETE strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas

Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.30	0.21	0.32	0.31	0.43	0.40
			[3.12]	[2.25]	[3.89]	[3.62]	[5.37]	[4.95]
Quintile	NYSE	EW	-0.10	-0.16	-0.11	-0.08	-0.09	-0.07
			[-1.97]	[-3.13]	[-2.26]	[-1.62]	[-2.06]	[-1.58]
Quintile	Name	VW	0.27	0.18	0.31	0.31	0.42	0.40
			[2.65]	[1.80]	[3.40]	[3.28]	[4.80]	[4.51]
Quintile	Cap	VW	0.14	0.03	0.16	0.17	0.28	0.26
			[1.62]	[0.41]	[2.23]	[2.24]	[3.96]	[3.75]
Decile	NYSE	VW	0.25	0.18	0.31	0.29	0.42	0.38
			[2.19]	[1.64]	[2.93]	[2.67]	[3.91]	[3.56]

Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas

Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.25	0.15	0.26	0.25	0.32	0.31
			[2.63]	[1.66]	[3.09]	[2.98]	[3.99]	[3.81]
Quintile	NYSE	EW	-0.32					
			[-5.18]					
Quintile	Name	VW	0.22	0.12	0.24	0.23	0.30	0.29
			[2.14]	[1.19]	[2.58]	[2.54]	[3.40]	[3.29]
Quintile	Cap	VW	0.11		0.10	0.10	0.17	0.16
			[1.18]		[1.35]	[1.40]	[2.42]	[2.36]
Decile	NYSE	VW	0.18	0.13	0.24	0.22	0.30	0.28
			[1.63]	[1.12]	[2.21]	[2.08]	[2.81]	[2.65]

Table 3: Conditional sort on size and ETE

This table presents results for conditional double sorts on size and ETE. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on ETE. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high ETE and short stocks with low ETE .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results

		ETE Quintiles					ETE Strategies					
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Size quintiles	(1)	0.95	0.90	0.90	0.82	0.71	-0.24	-0.26	-0.20	-0.16	-0.14	-0.11
		[3.47]	[4.36]	[4.26]	[3.46]	[2.67]	[-2.12]	[-2.32]	[-1.78]	[-1.38]	[-1.20]	[-0.93]
	(2)	0.82	0.86	0.85	0.86	0.76	-0.06	-0.14	-0.07	-0.05	-0.07	-0.05
		[3.67]	[4.37]	[4.11]	[3.77]	[3.07]	[-0.72]	[-1.74]	[-0.95]	[-0.59]	[-0.94]	[-0.68]
	(3)	0.72	0.74	0.83	0.82	0.78	0.06	-0.04	0.05	0.05	0.11	0.10
		[3.59]	[4.10]	[4.19]	[3.91]	[3.34]	[0.62]	[-0.39]	[0.53]	[0.54]	[1.28]	[1.18]
	(4)	0.65	0.71	0.78	0.72	0.79	0.14	0.07	0.15	0.12	0.21	0.18
		[3.33]	[4.18]	[4.36]	[3.66]	[3.64]	[1.51]	[0.78]	[1.78]	[1.40]	[2.53]	[2.10]
	(5)	0.46	0.59	0.56	0.57	0.67	0.21	0.10	0.24	0.24	0.37	0.36
		[2.88]	[3.83]	[3.51]	[3.26]	[3.51]	[1.89]	[0.90]	[2.47]	[2.43]	[3.96]	[3.73]

Panel B: Portfolio average number of firms and market capitalization

		ETE Quintiles Average n					ETE Quintiles Average market capitalization ($\$10^6$)				
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
Size quintiles	(1)	316	320	322	320	318	23	28	31	30	26
	(2)	102	102	102	102	102	51	53	53	53	52
	(3)	77	77	77	77	77	94	96	97	97	95
	(4)	66	66	66	66	66	207	213	209	210	205
	(5)	62	62	62	62	62	1352	2025	1600	1711	1467

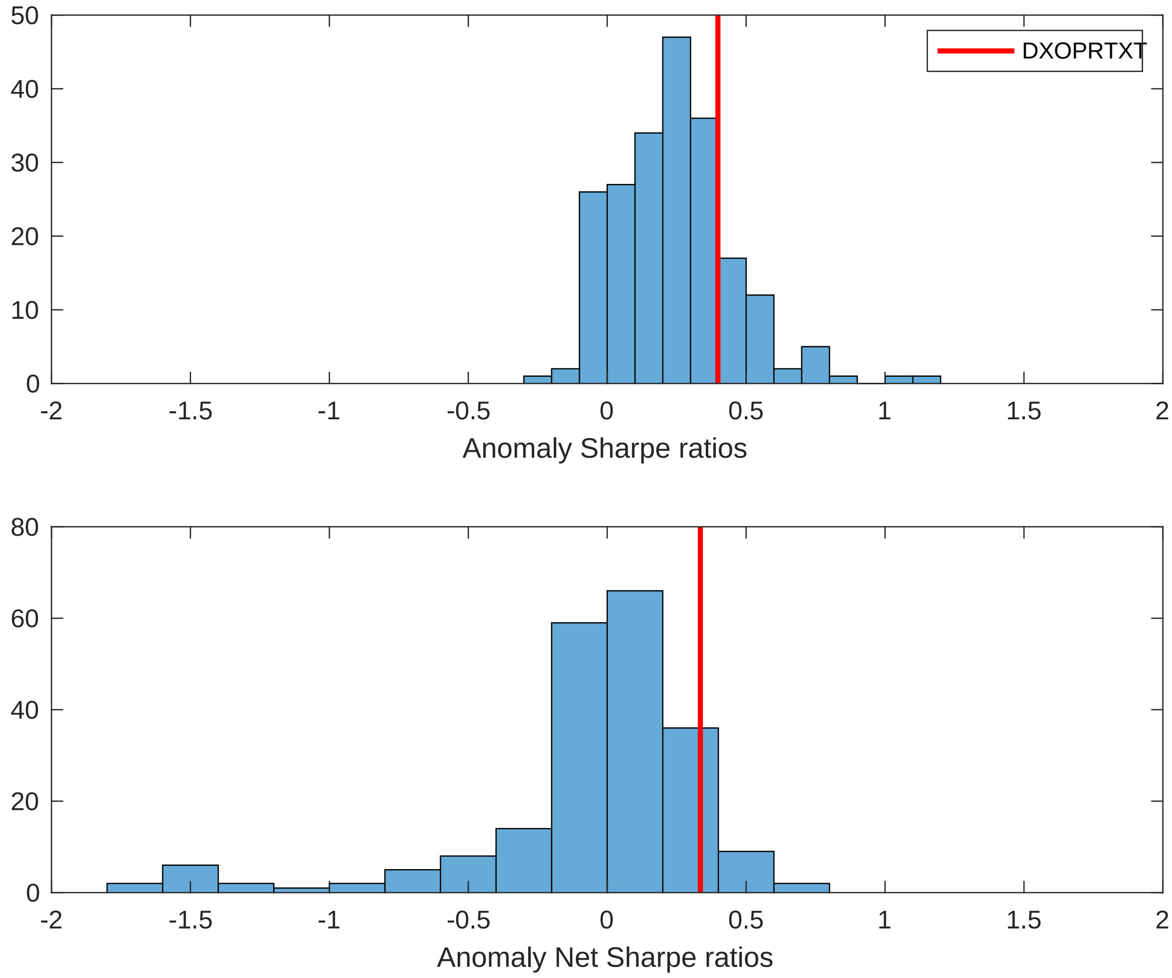


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the ETE with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

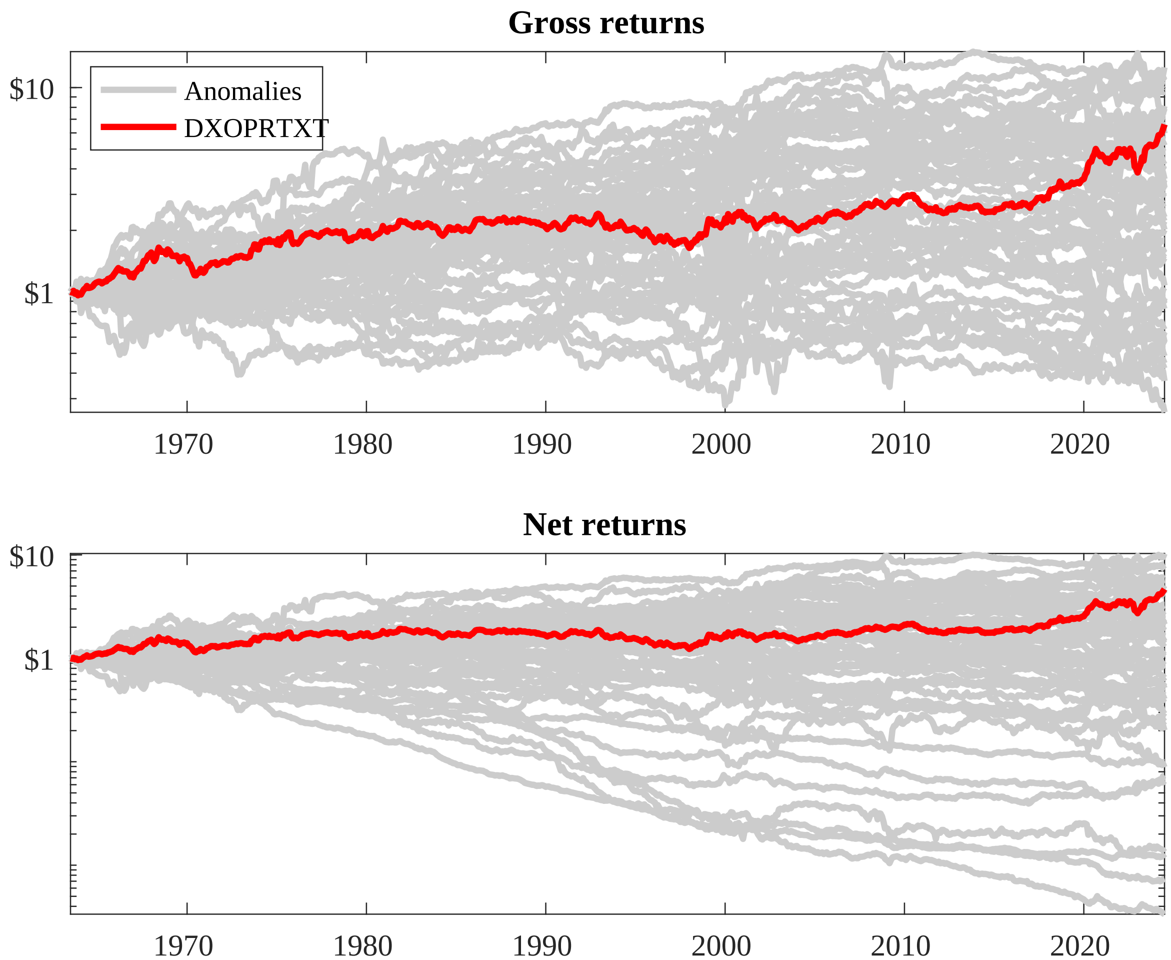
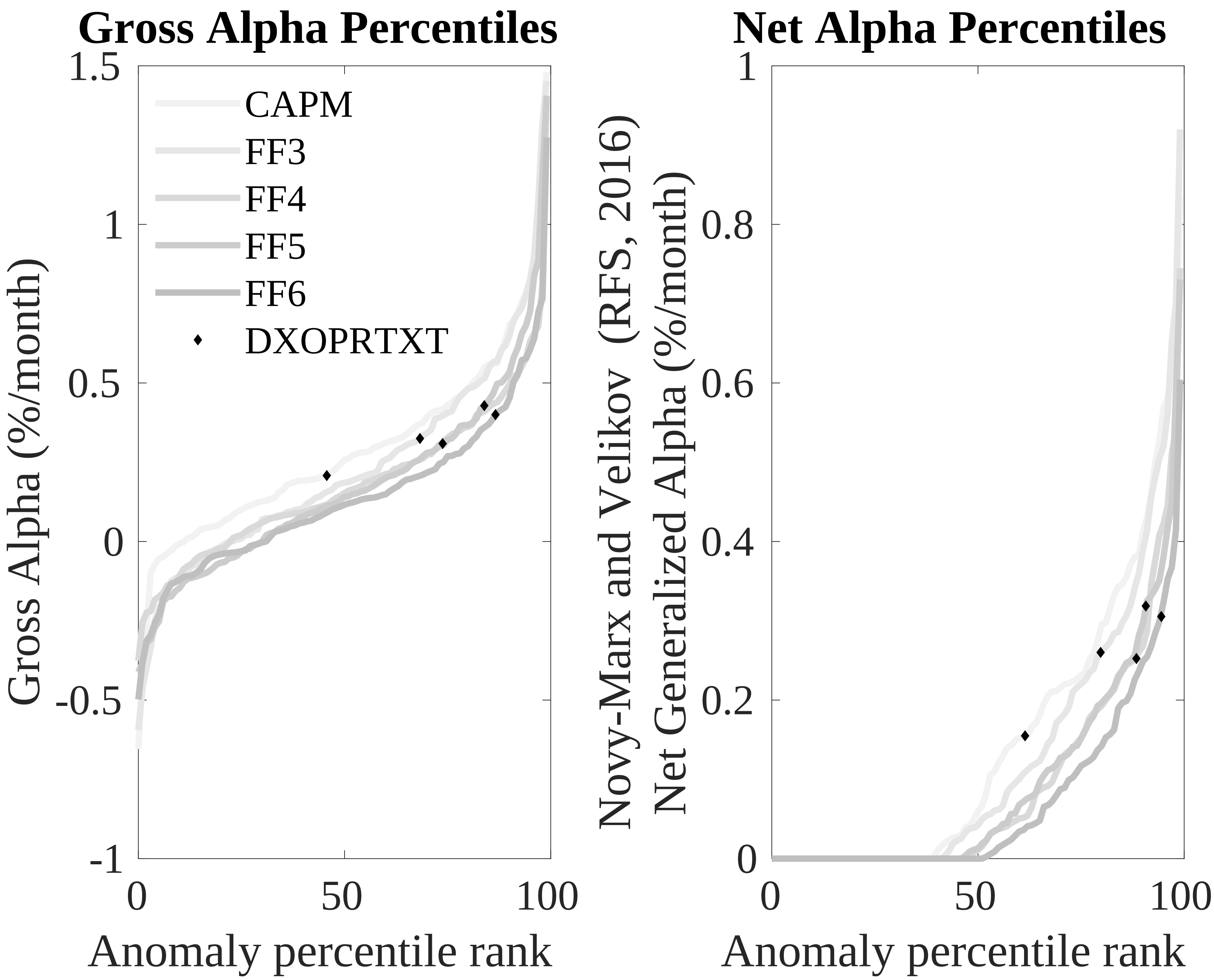


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the ETE trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Gross Alpha (%/month)
1.5
1
0.5
0
-0.5
-1
CAPM
FF3
FF4
FF5
FF6
DXOPRTXT
0
50
100
Anomaly percentile rank
Net Alpha Percentiles
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
1
0.8
0.6
0.4
0.2
0
0
50
100
Anomaly percentile rank

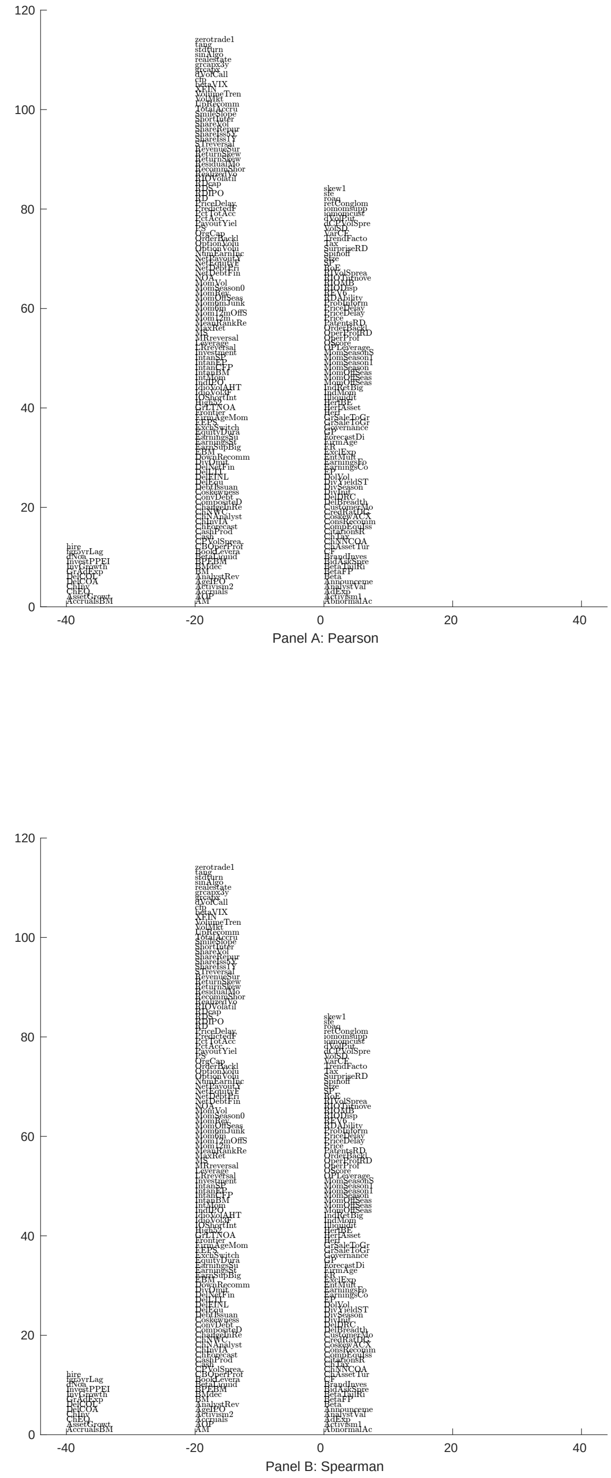


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with ETE. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

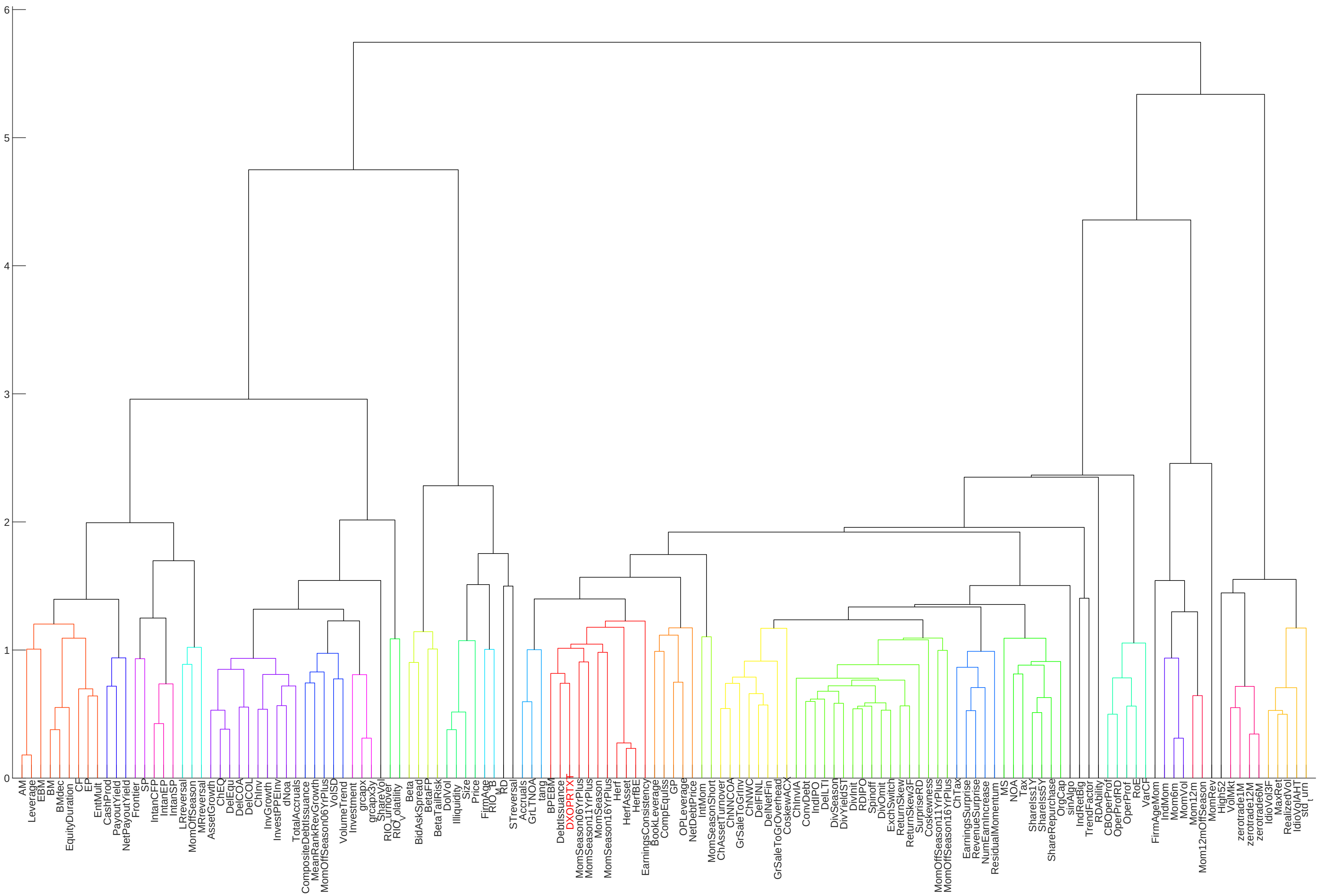


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

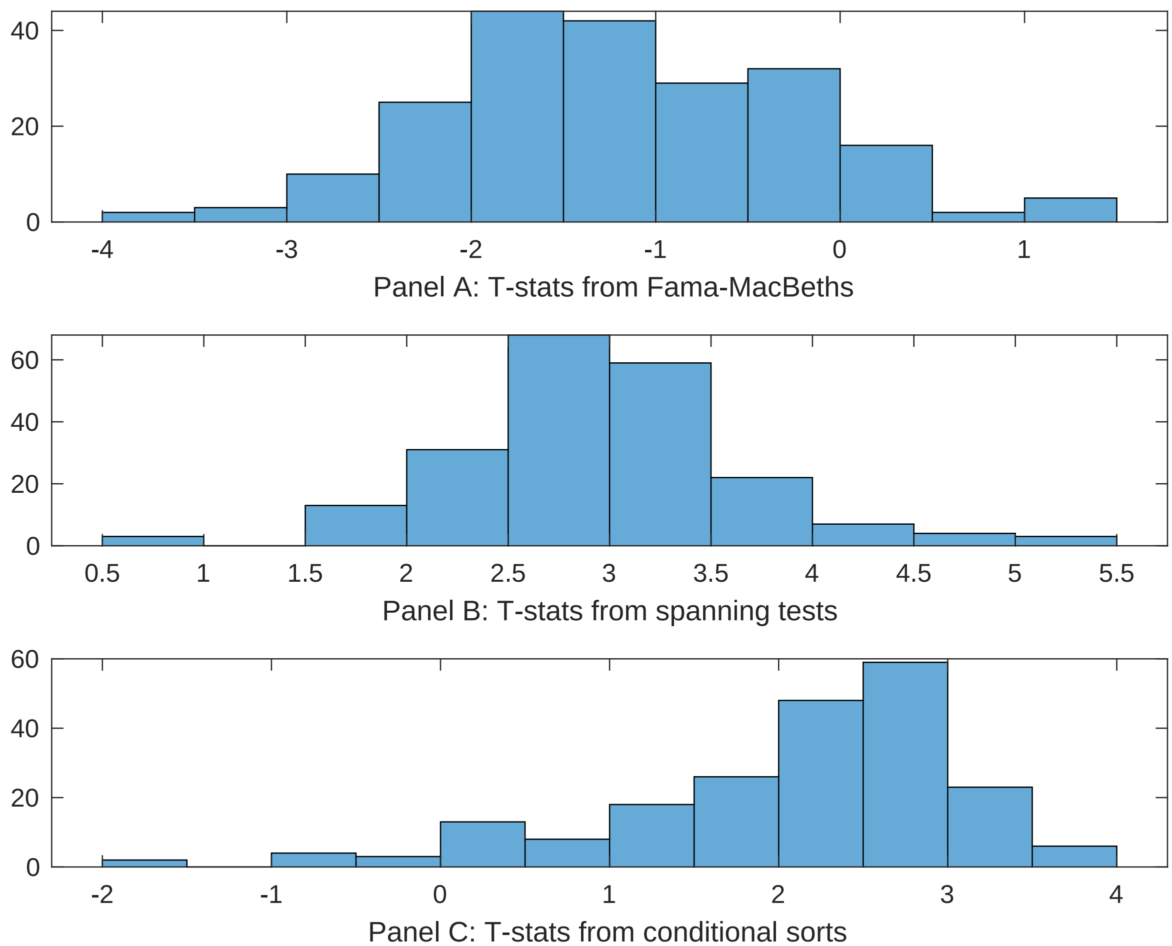


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of ETE conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ETE} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ETE} ETE_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ETE,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on ETE. Stocks are finally grouped into five ETE portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ETE trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies
This table presents Fama-MacBeth results of returns on ETE. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{ETE} ETE_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Change in current operating assets, Employment growth, Asset growth, Long-term EPS forecast, change in ppe and inv/assets, Change in current operating liabilities. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13	0.12	0.13	0.13	0.14	0.12	0.13
	[6.15]	[5.87]	[6.56]	[6.40]	[6.43]	[6.04]	[6.32]
ETE	-0.34	-0.72	-0.37	-0.57	0.42	-0.67	-0.18
	[-0.57]	[-1.11]	[-0.06]	[-1.53]	[0.06]	[-1.04]	[-0.48]
Anomaly 1	0.18						0.10
	[5.84]						[2.44]
Anomaly 2		0.83					-0.47
		[6.01]					[-0.03]
Anomaly 3			0.96				0.46
			[8.60]				[4.63]
Anomaly 4				0.76			-0.22
				[0.92]			[-0.03]
Anomaly 5					0.15		0.16
					[8.20]		[0.50]
Anomaly 6						0.14	-0.22
						[3.61]	[-3.87]
# months	732	713	732	504	732	732	504
$\bar{R}^2(\%)$	0	0	0	1	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the ETE trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{ETE} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Change in current operating assets, Employment growth, Asset growth, Long-term EPS forecast, change in ppe and inv/assets, Change in current operating liabilities. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.38	0.26	0.39	0.35	0.39	0.31	0.24
	[4.85]	[3.30]	[4.85]	[3.78]	[4.93]	[3.94]	[2.84]
Anomaly 1	-25.80						-8.53
	[-6.67]						[-1.77]
Anomaly 2		-42.09					-34.65
		[-10.05]					[-6.44]
Anomaly 3			-13.49				9.47
			[-2.80]				[1.87]
Anomaly 4				-12.99			-2.41
				[-4.13]			[-0.80]
Anomaly 5					-25.12		-19.72
					[-6.80]		[-4.88]
Anomaly 6						-24.96	-7.20
						[-6.67]	[-1.45]
mkt	2.71	1.41	2.52	-3.95	2.85	3.44	-1.11
	[1.44]	[0.77]	[1.31]	[-1.63]	[1.51]	[1.82]	[-0.51]
smb	8.99	12.47	15.45	3.59	15.37	13.63	2.65
	[3.16]	[4.72]	[5.48]	[1.05]	[5.64]	[5.00]	[0.81]
hml	0.91	-1.82	-8.35	-13.49	-6.04	0.83	-2.89
	[0.23]	[-0.50]	[-2.23]	[-3.07]	[-1.65]	[0.21]	[-0.66]
rmw	-1.30	0.99	1.91	11.00	1.86	3.70	6.27
	[-0.35]	[0.28]	[0.51]	[2.55]	[0.51]	[1.00]	[1.60]
cma	-44.71	-21.91	-42.58	-45.74	-39.59	-44.88	-11.74
	[-7.76]	[-3.44]	[-5.32]	[-6.94]	[-6.54]	[-7.80]	[-1.40]
umd	2.87	5.96	3.53	4.63	4.00	2.34	4.67
	[1.53]	[3.27]	[1.84]	[2.19]	[2.14]	[1.24]	[2.43]
# months	732	713	732	504	732	732	504
$\bar{R}^2$(%)	39	44	36	44	40	39	55

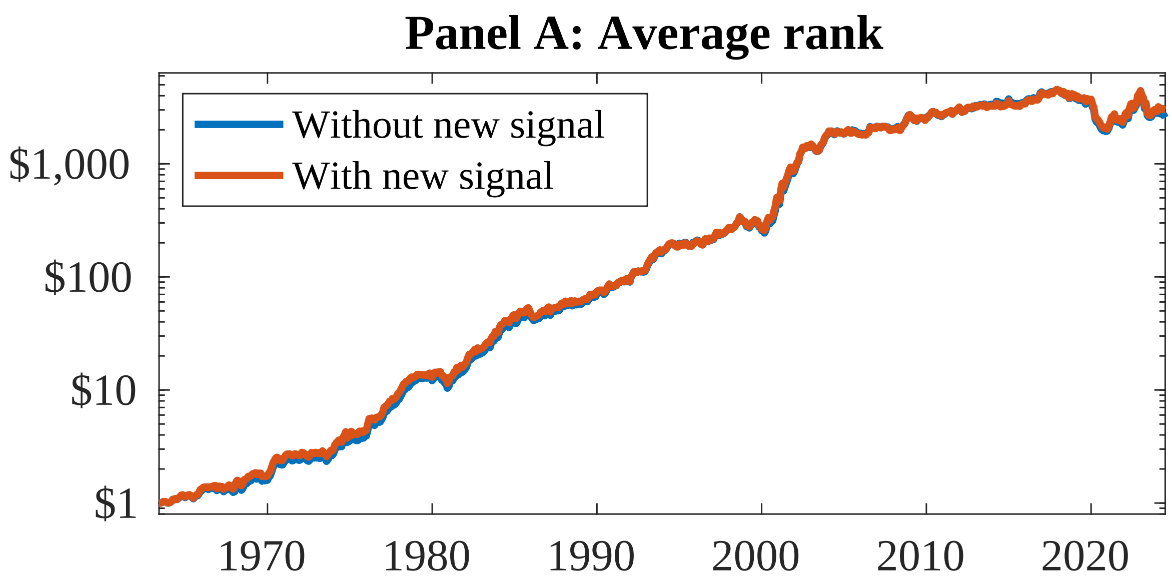


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as ETE. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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